

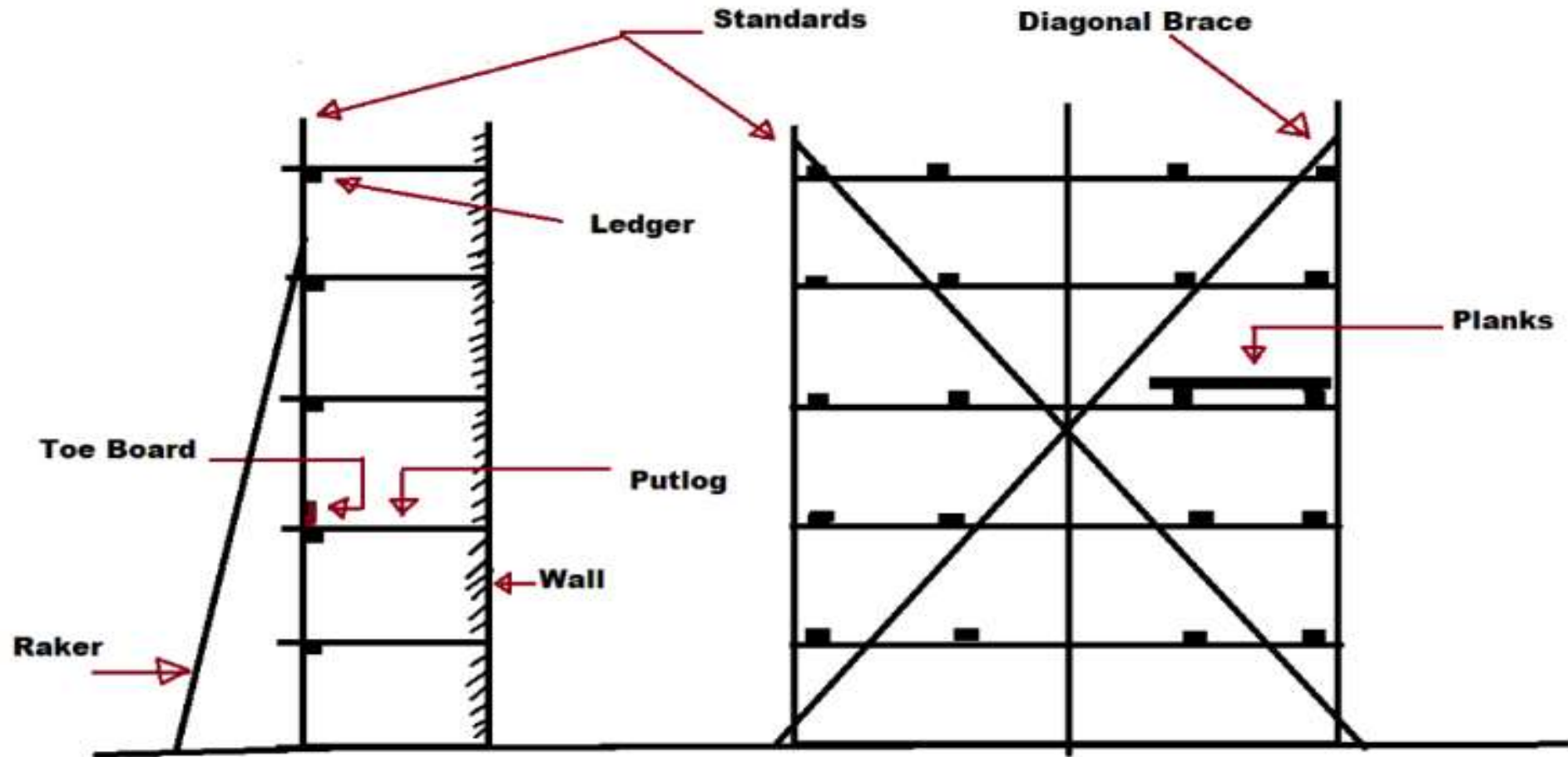
Scaffolding and Its Types

Scaffolding:

- A temporary structure used to support a work crew and materials to aid in the construction, maintenance and repair of buildings, bridges and all other man-made structures.
- Scaffolds are widely used on site to get access to heights and areas that would be otherwise hard to get to.
- Scaffolding is a temporary structure that is typically made from metal poles and wood planks and used to support construction workers, inspectors, cleaners, and others who need to work at height.
- Scaffoldings provides a temporary safe working platform for maintenance, construction, repair, access, inspection etc.



Part of scaffolding:



- **Standards-** Standards is a vertical member of scaffolding, which may be vertical bullies or poles or other vertically supported members. In the end, all the loads are carried by standards. So, they should be strong and rigid enough to bear them.
- **Ledgers-** Ledgers are the horizontal members of scaffolding, which are laid parallel to the walls. Ledgers are fixed with standards either by ropes or by using other connectors. The connection between Ledgers and standards should be as possible as rigidly, otherwise, it may cause an accident due to loose connection between them

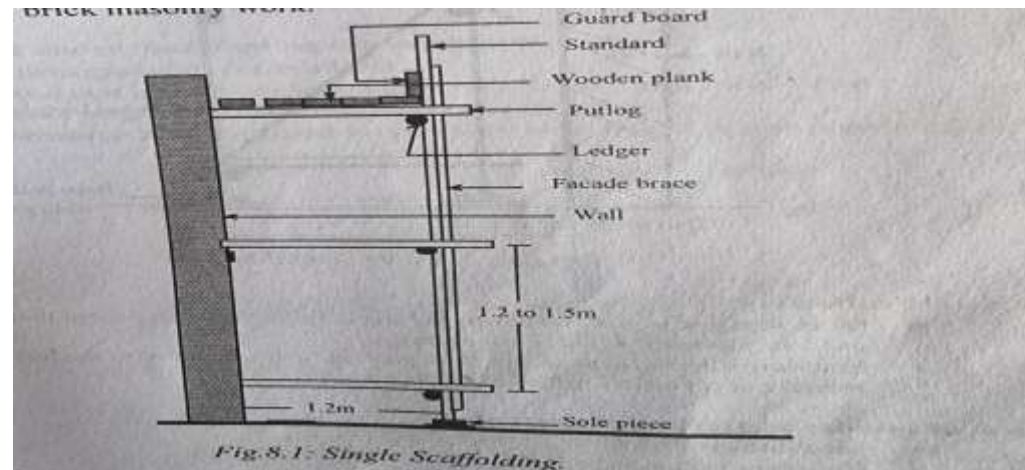
- **Putlogs-** Putlogs are also a horizontal member, but it is at right angle to the wall and standards. One end of these members are laid on Ledgers, and other end is held in the wall.
- **Braces-** These are the members that are set diagonally on the standards to provide stiffness to the framework.
- **Planks or Boarding-** Planks or boarding is a horizontal member made of wood, on which workers perform their particular work. For a particular work, all the required materials and appliances are kept over it.

- **Transoms-** These are those putlogs whose both ends are supported on ledgers.
- **Guard Rail-** This is a rail provided like ledger at the working level.
- **Toe Board-** These are boards, placed parallel to ledgers, support on put log to give protection at the level of working plat form.

Types of Scaffolding:

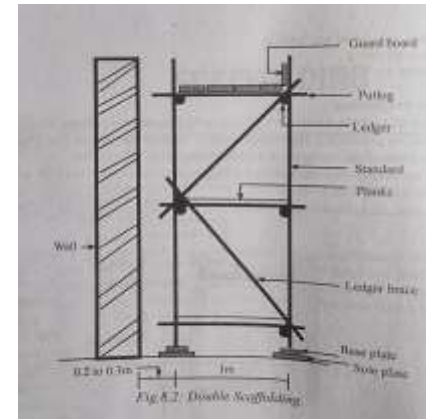
1. Single Scaffolding or brick layers Scaffolding

- Single scaffolding is generally used for brick masonry and is also called as brick layer's scaffolding.
- Single scaffolding consists of standards, ledgers, putlogs etc., which is parallel to the wall at a distance of about 1.2 m.
- Distance between the standards is about 2 to 2.5 m.



2. Double Scaffolding

- Double Scaffolding is generally used for stone masonry so, it is also called as mason's scaffolding.
- In stone walls, it is hard to make holes in the wall to support putlogs. So, two rows of scaffolding is constructed to make it strong.
- The first row is 20 – 30 cm away from the wall and the other one is 1m away from the first row. Widely used for stone masonry.



3. Steel Scaffolding

- Steel scaffolding is constructed by steel tubes which are fixed together by steel couplers or fittings.
- It is very easy to construct or dismantle.
- It has greater strength, greater durability and higher fire resistance.
- It is not economical but will give more safety for workers. So, it is used extensively nowadays.



4. Needle or Cantilever Scaffolding

- This is a type of scaffolding in which the standards are supported on a series of needles and these needles are taken out through holes in the wall.
- This is called single frame type scaffolding. In the other type needles are strutted inside the floors through the openings and this is called independent or double frame type scaffolding.
- Care should be taken while construction of cantilever scaffolding.



5. Trestle Scaffolding

- In Trestle scaffolding, the working platform is supported on movable tripods or ladders.
- This is generally used for work inside the room, such as paintings, repairs etc., up to a height of 5m.



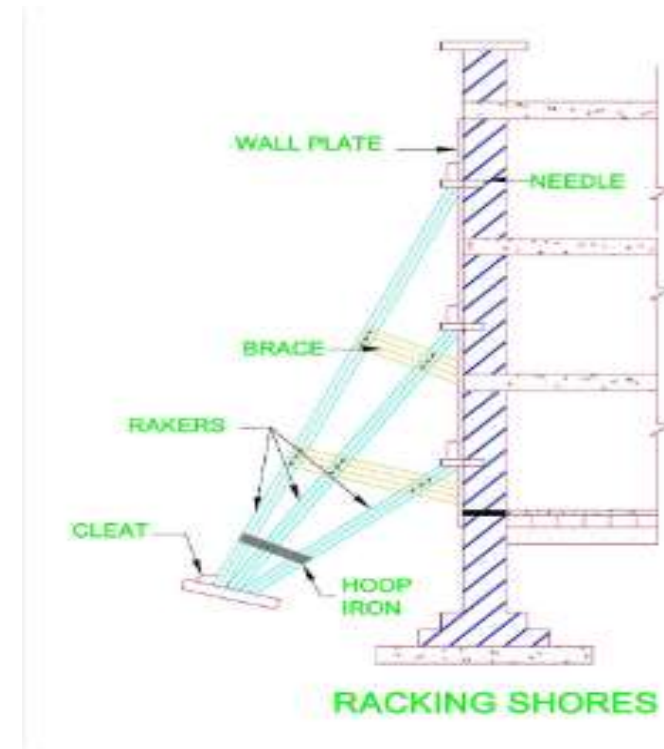
6. Suspended Scaffolding

- In suspended scaffolding, the working platform is suspended from roofs with the help of wire ropes or chains etc.,
- It can be raised or lowered to our required level.
- This type of scaffolding is used for repair works, pointing, paintings etc.



Shoring:

- Shoring is the practice of building a temporary structure to support an unsafe structure on construction and excavation sites.
- Shoring is the construction of a temporary structure to support temporarily an unsafe structure.



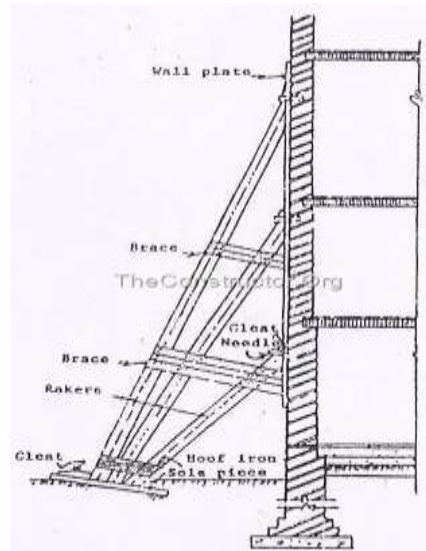
Objective of shoring:

- Defective parts of the building are to be dismantled and rebuilt.
- Support upper part of the wall during formation of large openings on the wall.
- Avoid failure of structure while removing adjoining structure.
- Give support to wall of the building develops sign of bulging.
- Intermediate building is to be dismantled.

Types of Shoring

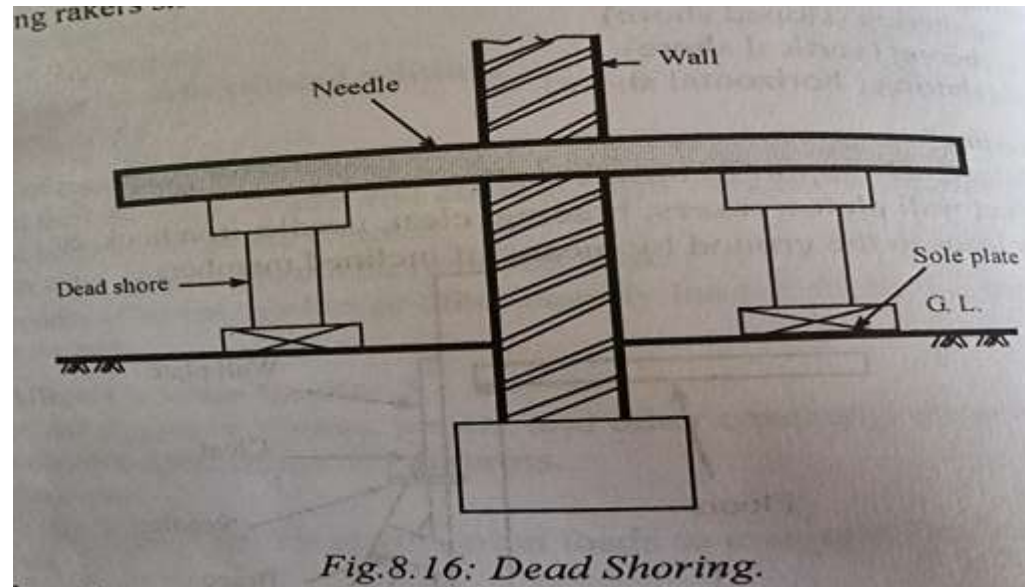
a) Raking Shoring (sloped shore)

- Provide inclined support to unsafe wall.
- Consists of wall plates, rakers, bracing, cleat, needle, iron hook etc.
- Transfer load to the ground by means of inclined members.



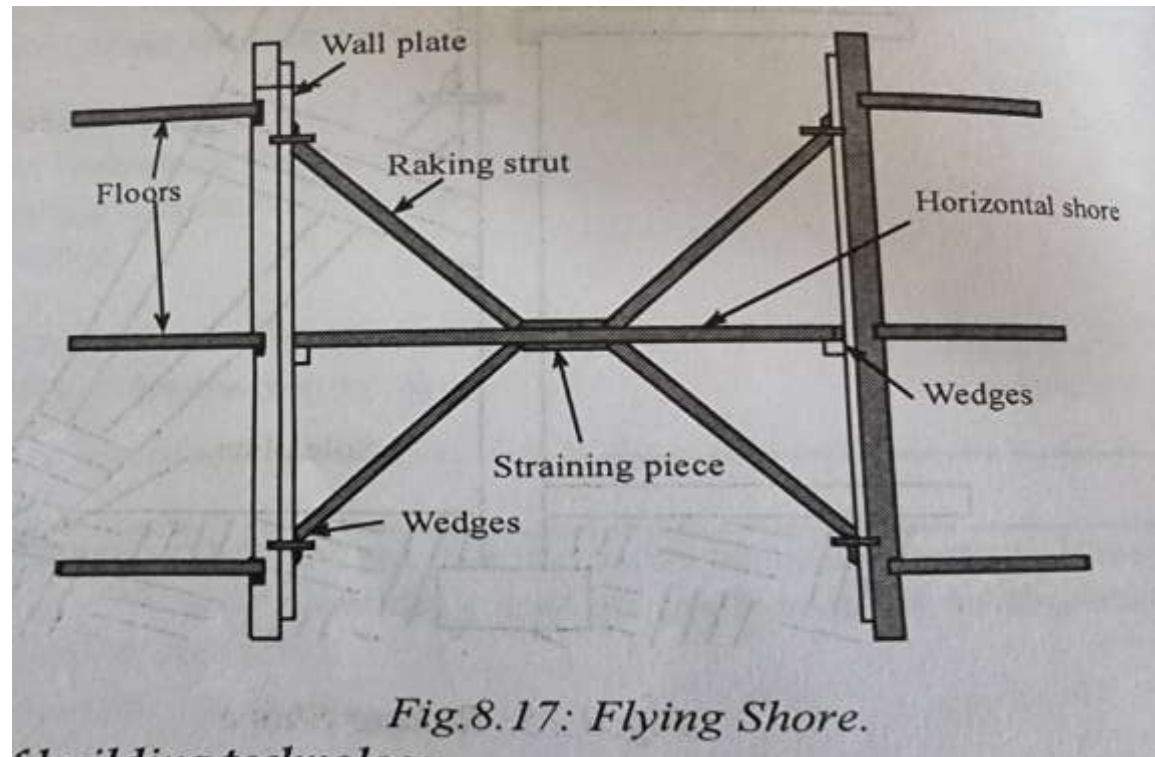
b) Dead Shoring (Vertical shore)

- Dead shore is the system of shoring which is used to render vertical support to walls and roofs, floors, etc.
- Vertical members are used to support dead load.
- Used in vertical wall, floors and roofs when the lower part of building has to be removed, strengthen etc.



c) Flying Shoring (horizontal shore)

- Horizontal members are used to support load.
- Provide clear working distance under the shoring.



Underpinning:

- Underpinning is the process of strengthening the foundation of an existing building or other structure.
- Underpinning is the reinforcing of an existing building foundation.
- It is required when the original foundation is no longer strong enough to support the house.

Need of underpinning

- Basement has to be provided below existing foundation.
- Settlement in the foundation and serious cracks in wall.
- The original foundation is not strong or stable enough.
- To support a structure that is sinking or tilting due to poor soil.
- Earthquake, flood or other natural causes have caused the structure to move, thereby requiring stabilization of foundation soils or footing.

Method of underpinning

a) Pit method:

- In this method, the entire length of the foundation to be underpinned is divided into sections of 1.2 to 1.5 m lengths. One section is taken up at a time.
- A hole is made in the wall for all divided sections above the plinth level, and needle is inserted in the hole.
- The needle may be made of a material of stout, timber or steel section.
- Bearing plates are placed above the needle to support the masonry above it.

- The supporting arrangement of the needle is made by crib supports (wooden blocks) on both sides of the wall and screw jacks.
- After that, a foundation pit is excavated up to the required level of new foundation.
- Then the new foundation is laid in the pit.
- When the work of one section is over, work on next to next section is taken up, i.e., alternate sections are underpinned in the first round, and then the remaining sections are taken up.

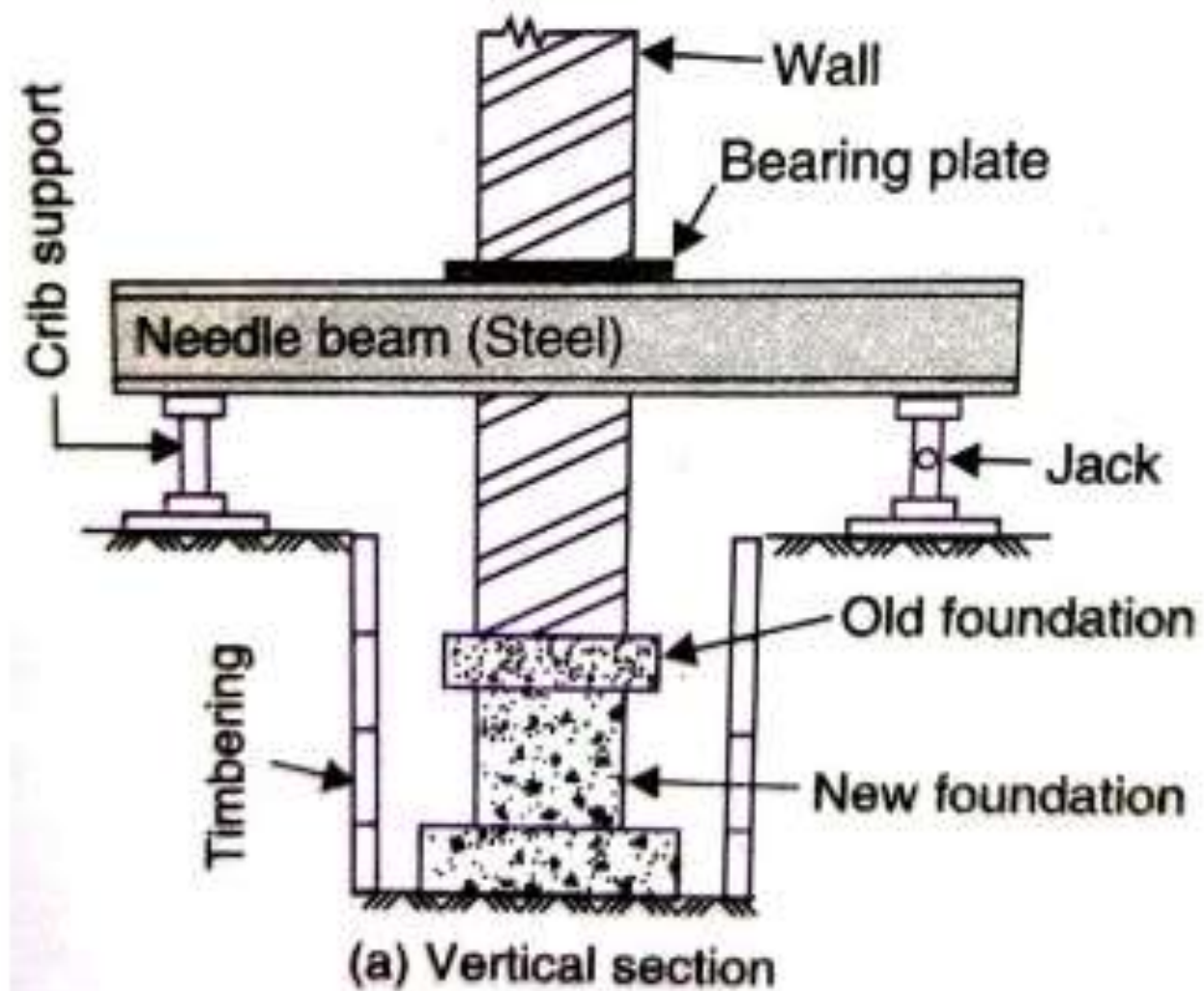


Fig. 2. Pit Method of Underpinning

b) Pile method:

- In the Pile method of underpinning, as the name suggests Piles are installed by proper driving technique along both sides of the wall to be strengthened. The piling technique normally used are borehole pile on under-reamed piles are used.
- After that, concrete or steel needles are penetrated through the wall and are connected to the Pile. These needles function is as beams and is act as pile caps also.
- Pile method is suitable for clayey soils, waterlogged areas, and also in weak bearing strata.

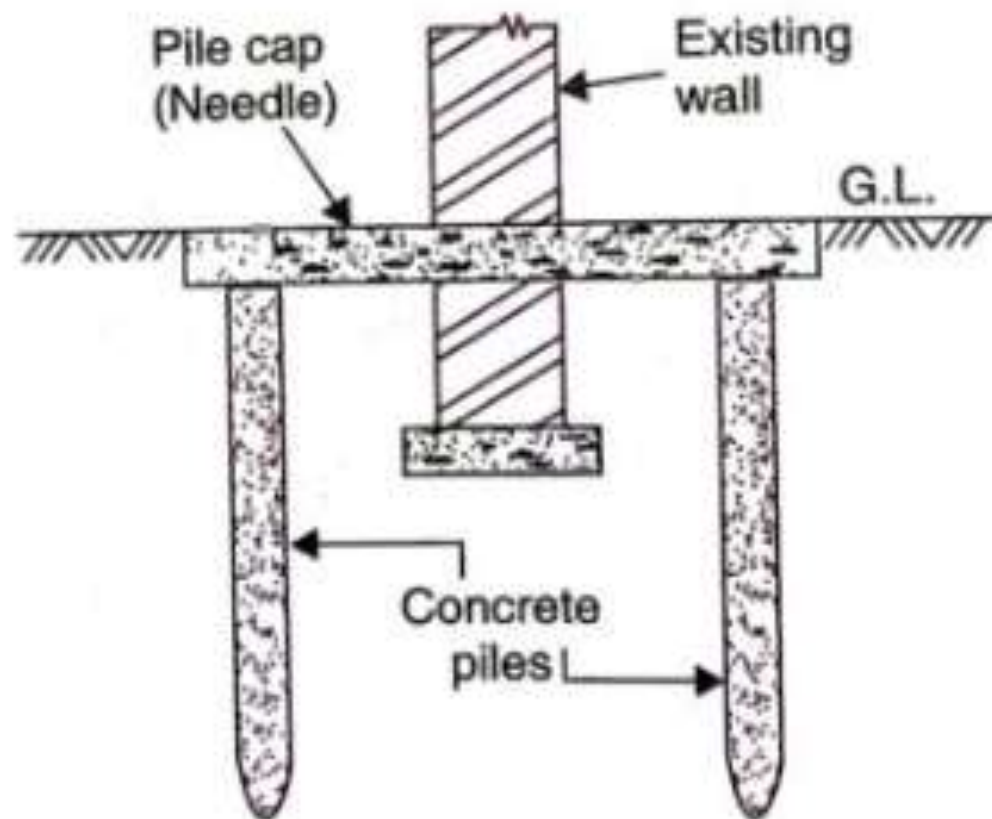


Fig. 3. Pile Method of Underpinning

Fire Fighting Techniques and Fire Extinguishing Equipment:

Fire Fighting Techniques:

1. Cooling

- Reduces the temperature of the fire below its ignition point.
- Water is commonly used for cooling.

2. Smothering

- Cuts off the oxygen supply to the fire.
- Carbon dioxide (CO₂), and fire blankets are used.

3. Starvation

- Removes or isolates the fuel source.
- Example: Turning off gas or fuel supply.

4. Breaking the Chain Reaction

- Interrupts the chemical reaction occurring during combustion.
- Dry chemical powder extinguishers are commonly used.

Fire Extinguishing Equipment:

1. Water Fire Extinguisher

- Used for Class A fires (wood, paper, cloth).
- Not suitable for electrical or oil fires.

2. Foam Fire Extinguisher

- Used for Class A and Class B fires (flammable liquids).
- Forms a blanket over the fuel surface.

3. Carbon Dioxide (CO₂) Fire Extinguisher

- Suitable for electrical fires and flammable liquids.
- Leaves no residue after use.

4. Dry Chemical Powder (DCP) Extinguisher

- Effective for Class A, B, C, and electrical fires.
- Interrupts the combustion process.

5. Wet Chemical Extinguisher

- Used for cooking oil and fat fires (Class K/F).
- Commonly found in kitchens.

6. Fire Blanket

- Used to smother small fires.
- Suitable for kitchen and clothing fires.

7. Fire Hose Reel

- Provides a continuous supply of water for firefighting.

8. Automatic Sprinkler System

- Automatically sprays water when high temperatures are detected.
- Commonly installed in commercial and residential buildings.

Fire Class	Materials Involved
Class A	Wood, paper, cloth, rubber
Class B	Petrol, diesel, paint, oil
Class C	Flammable gases (LPG, propane)
Class D	Combustible metals
Class (K,F)	Cooking oils and fats
Electrical	Energized electrical equipment



CARBON DIOXIDE FIRE EXTINGUISHER 2KG CARBON DIOXIDE

- 1 USE UPRIGHT PULL OUT PIN
- 2 AIM HORN AT BASE OF FIRE
- 3 SQUEEZE HANDLE TO OPERATE
RELEASE HANDLE TO STOP



CAUTION: DO NOT HOLD HORN WHEN OPERATING.
WARNING: THIS IS A PRESSURISED CONTAINER.

RECHARGE AFTER ANY USE.
THE EXTINGUISHER SHOULD BE SERVICED AT LEAST ONCE A YEAR BY A
QUALIFIED COMPETENT PERSON IN ACCORDANCE WITH BS 5306.
CHARGING SHOULD BE CARRIED OUT TO BS 5445 PART 1 ONLY USING
COMPONENTS RECOMMENDED BY THE MANUFACTURER.

FOAM FIRE EXTINGUISHER 6L FOAM

- 1 USE UPRIGHT PULL OUT PIN
- 2 AIM HOSE AT BASE OF FIRE
- 3 SQUEEZE HANDLE TO OPERATE
RELEASE HANDLE TO STOP



CAUTION: IF PRESSURE INDICATOR ENTERS RED
SECTION OF GAUGE WITHDRAW FROM SERVICE.
DO NOT USE OUTSIDE TEMPERATURE RANGE.
WARNING: THIS IS A PRESSURISED CONTAINER.

RECHARGE AFTER ANY USE. PROTECT FROM FROST.
EXTINGUISHER SHOULD BE SERVICED AT LEAST ONCE A YEAR BY A
QUALIFIED COMPETENT PERSON IN ACCORDANCE WITH BS 5306.
CHARGING SHOULD BE CARRIED OUT TO BS 5445 PART 1 ONLY USING
COMPONENTS RECOMMENDED BY THE MANUFACTURER.

EXTINGUISHING MEDIUM: 91.7% WATER + 8.3% JOE FOAM
OPERATING TEMPERATURE RANGE: +5°C TO +55°C
TEST PRESSURE: 30BAR
WORKING PRESSURE: 12BAR
PRESSURISING GAS: NITROGEN

POWDER FIRE EXTINGUISHER 9KG ABC POWDER

- 1 USE UPRIGHT PULL OUT PIN
- 2 AIM HOSE AT BASE OF FIRE
- 3 SQUEEZE HANDLE TO OPERATE
RELEASE HANDLE TO STOP



CAUTION: IF PRESSURE INDICATOR ENTERS RED
SECTION OF GAUGE WITHDRAW FROM SERVICE.
DO NOT USE OUTSIDE TEMPERATURE RANGE.
WARNING: THIS IS A PRESSURISED CONTAINER.

RECHARGE AFTER ANY USE.
EXTINGUISHER SHOULD BE SERVICED AT LEAST ONCE A YEAR BY A
QUALIFIED COMPETENT PERSON IN ACCORDANCE WITH BS 5306.
CHARGING SHOULD BE CARRIED OUT TO BS 5445 PART 1 ONLY USING
COMPONENTS RECOMMENDED BY THE MANUFACTURER.

EXTINGUISHING MEDIUM: ABC POWDER
OPERATING TEMPERATURE RANGE: -50°C TO +60°C
TEST PRESSURE: 30BAR
WORKING PRESSURE: 14BAR
PRESSURISING GAS: NITROGEN

WATER FIRE EXTINGUISHER 9L WATER

- 1 USE UPRIGHT PULL OUT PIN
- 2 AIM HOSE AT BASE OF FIRE
- 3 SQUEEZE HANDLE TO OPERATE
RELEASE HANDLE TO STOP



CAUTION: IF PRESSURE INDICATOR ENTERS RED
SECTION OF GAUGE WITHDRAW FROM SERVICE.
DO NOT USE OUTSIDE TEMPERATURE RANGE.
WARNING: DO NOT USE ON LIVE ELECTRIC EQUIPMENT.

RECHARGE AFTER ANY USE. PROTECT FROM FROST.
EXTINGUISHER SHOULD BE SERVICED AT LEAST ONCE A YEAR BY A
QUALIFIED COMPETENT PERSON IN ACCORDANCE WITH BS 5306.
CHARGING SHOULD BE CARRIED OUT TO BS 5445 PART 1 ONLY USING
COMPONENTS RECOMMENDED BY THE MANUFACTURER.

EXTINGUISHING MEDIUM: WATER
OPERATING TEMPERATURE RANGE: +5°C TO +60°C
TEST PRESSURE: 30BAR
WORKING PRESSURE: 10BAR
PRESSURISING GAS: NITROGEN

Repair and Strengthening of Structures:

Repair:

- Process of restoring damaged structural elements to their original condition.
- Improves serviceability and appearance.

Strengthening:

- Increases the load-carrying capacity of an existing structure.
- Enhances durability and safety.

Causes of Cracks in Buildings:

Structural Causes:

- Overloading
- Earthquake
- Design errors
- Poor construction quality

Non-Structural Causes

- Temperature variation
- Shrinkage
- Moisture movement
- Poor workmanship

Structural Cracks:

- Cracks that affect the strength and stability of a structure.

Characteristics:

- Wide cracks
- Extend through structural members
- Continue to grow over time

Common Locations:

- Beams
- Columns
- Slabs
- Foundations

Effects:

- Reduced strength
- Water leakage
- Corrosion
- Structural failure

Non-Structural Cracks:

- Cracks that affect only appearance and serviceability.

Examples

- Plaster cracks
- Hairline wall cracks
- Shrinkage cracks

Effects

- Poor appearance
- Water seepage
- Reduced durability

Types of Structural Cracks

- Flexural cracks
- Shear cracks
- Torsional cracks
- Settlement cracks
- Corrosion cracks
- Earthquake-induced cracks

Types of Non-Structural Cracks

- Plastic shrinkage cracks
- Drying shrinkage cracks
- Thermal cracks
- Expansion and contraction cracks
- Plaster cracks

Repair Techniques:

Surface Repair

- Cement mortar
- Polymer mortar

Epoxy Injection

- Used for narrow structural cracks

Routing and Sealing

- Suitable for non-structural cracks

Stitching

- Steel staples across cracks

Grouting

- Cement grout
- Chemical grout

Strengthening Techniques

- Concrete jacketing
- Steel jacketing
- Fiber Reinforced Polymer (FRP) wrapping
- Section enlargement
- External prestressing
- Plate bonding

Retrofitting

- Retrofitting is the process of modifying an existing structure to improve its strength, stability, and performance against future loads such as earthquakes.

Objectives

- Increase strength
- Improve ductility
- Improve seismic performance
- Increase service life

Need for Retrofitting

- Earthquake resistance
- Increased load requirements
- Aging structures
- Corrosion damage
- Design deficiencies
- Change in building usage

Retrofitting Techniques

❖ Concrete Jacketing

- Additional reinforced concrete layer around members.

❖ Steel Jacketing

- Steel plates or angles around columns and beams.

❖ Fiber Reinforced Polymer (FRP) wrapping

- Carbon Fiber Reinforced Polymer (CFRP)
- Glass Fiber Reinforced Polymer (GFRP)

❖ **Plate Bonding**

- Steel plates bonded using epoxy.

❖ **External Prestressing**

- External tendons increase strength.

Advantages of Retrofitting

- Cost-effective
- Improves earthquake resistance
- Extends service life
- Increases structural strength
- Less demolition required
- Faster than reconstruction

Limitations

- Requires skilled labor
- High material cost
- Difficult inspection
- Temporary interruption of building use
- Proper design required

Structural Cracks	Non-Structural Cracks
Affect safety	Affect appearance
Require immediate repair	Usually cosmetic
Deep cracks	Hairline cracks
Occur in beams, columns, slabs	Occur in plaster and walls
Reduce strength	Do not reduce strength significantly

Demolition of Structures:

- Demolition is the systematic dismantling or destruction of a building or structure in a safe and controlled manner.
- It is carried out when a structure is unsafe, obsolete, or needs to be replaced.
- Demolition is the safe and planned removal of structures.
- Proper planning and legal approval are essential.
- Safety is the highest priority.
- Various demolition methods are selected based on building size, location, and structural type.
- Recycling demolition waste supports sustainable construction.

Objectives

- Remove unsafe structures.
- Clear land for new construction.
- Ensure public and worker safety.
- Recover reusable materials through recycling.

Reasons for Demolition

- Building has become structurally unsafe.
- Severe damage due to earthquake, fire, or flood.
- Building has exceeded its service life.
- Redevelopment or urban renewal projects.
- Change in land use.
- Illegal or unauthorized construction.
- High maintenance costs.

Steps Before Demolition

Before demolition begins, the following steps should be completed:

- Structural inspection and condition assessment.
- Obtain legal approvals and permits.
- Prepare a demolition plan.
- Disconnect electricity, water, gas, and communication services.
- Identify and remove hazardous materials (e.g., asbestos).
- Install barricades, warning signs, and safety measures.
- Inform nearby residents and authorities.
- Arrange waste disposal and recycling plans.

Safety Precautions

- Workers should wear Personal Protective Equipment (PPE).
- Install fencing around the demolition site.
- Control dust using water spraying.
- Prevent falling debris with safety nets.
- Maintain emergency exits and first-aid facilities.
- Monitor noise and vibration levels.
- Restrict unauthorized entry.

Demolition Planning

- A demolition plan should include:
- Building drawings and structural details.
- Sequence of demolition.
- Equipment and machinery required.
- Safety measures.
- Waste management plan.
- Emergency response procedures.
- Environmental protection measures.

Methods of Demolition

Common Methods

- Manual Demolition
- Mechanical Demolition
- Implosion (Controlled Explosive Demolition)
- High-Reach Arm Demolition
- Crane and Wrecking Ball Method
- Diamond Wire Sawing and Cutting

Manual Demolition

- Performed using hand tools such as hammers, crowbars, chisels, and jackhammers.

Suitable For

- Small buildings.
- Partial demolition.
- Interior dismantling.

Advantages

- Low equipment cost.
- Better control.
- Less vibration.

Disadvantages

- Time-consuming.
- Labor-intensive.
- Suitable only for small structures.

Mechanical Demolition

Equipment Used:

- Excavators
- Hydraulic breakers
- Crushers
- Bulldozers
- Loaders

Advantages

- Fast and efficient.
- Suitable for large buildings.
- Reduces labor requirements.

Disadvantages

- Expensive equipment.
- High noise and dust.
- Requires skilled operators.

Implosion Method

- Controlled use of explosives to collapse a structure inward.

Suitable For

- High-rise buildings.
- Large industrial structures.

Advantages

- Very fast.
- Efficient for tall structures.

Disadvantages

- Requires expert planning.
- High safety risks.
- Generates vibration and dust.

High-Reach Arm Demolition

- Uses long-boom excavators equipped with demolition attachments.

Applications

- Medium and high-rise buildings.

Advantages

- Safe operation from ground level.
- High efficiency.
- Controlled demolition.

Crane and Wrecking Ball Method

- A heavy steel ball suspended from a crane is used to break down walls.

Applications

- Large masonry and concrete structures.

Advantages

- Simple technique.
- Effective for large buildings.

Disadvantages

- Difficult to control.
- Generates excessive dust and vibration.
- Rarely used today due to safety concerns.

Selective Demolition

- Selective demolition involves removing only specific parts of a building while preserving the remaining structure.

Applications

- Building renovation.
- Remodeling.
- Interior modifications.

Benefits

- Less waste generation.
- Material recycling.
- Reduced environmental impact.

Waste Management After Demolition

- Materials that can be recycled:
- Concrete
- Steel
- Bricks
- Timber
- Glass
- Aluminum
- Plastic

Benefits

- Reduces landfill waste.
- Saves natural resources.
- Lowers construction costs.
- Supports sustainable construction.

Thank You